Design to Cost · FORMING SIMULATION REPORT (SHEET METAL)

EX26-0001 · L mounting bracket · rev A

Result: 1 CORRECTION(S) REQUIRED

• Inside bend radius 0.50 mm is below 1 × thickness (2.00 mm): cracking risk. Increase the radius.

Formability checks

Check	Result	Detail
bend radius	FAIL	Inside bend radius 0.50 mm is below 1 × thickness (2.00 mm): cracking risk. Increase the radius.
hole diameter	OK	passed (rule: hole_count > 0 and min_hole_d < min_hole_factor * t)
draw ratio	OK	passed (rule: draw_ratio > ldr)
springback	OK	passed (rule: springback_deg > max_springback_deg)

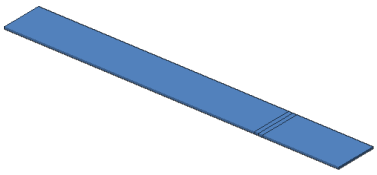
Part and blank

Item	Value
Material / thickness	Galvanised steel GI · 2 mm
Flat pattern	326.82 × 40.00 mm, 6 hole(s), 1 bend(s)
Blank (costing)	327.4 × 40.0 mm · Developed straight strip + bending stage
Inside bend radius	0.50 mm (minimum 1 × t)
Smallest hole	8.50 mm
Blank option: Rectangle keeping the formed width	250.0 × 80.0 mm · yield 65.5 %
Blank option: Developed straight strip + bending stage	327.4 × 40.0 mm · yield 100.0 %

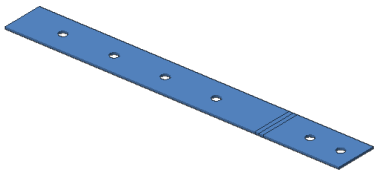
Stages

#	Operation	Force (t)	Press
1	Blanking	49.5	PRESS-100
2	Piercing	10.8	PRESS-100
3	Bending (V)	0.5	PRESS-100

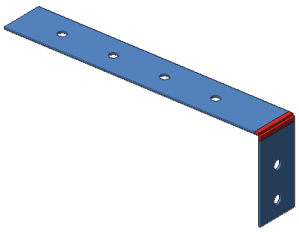
Part after every stage (red = bend that fails the bend-radius check)



After Blanking



After Piercing



After Bending (V)

Flat pattern with bend lines

Method: analytical checks and forces from the formulas in Master data (Sheet checks, Sheet operations, Sheet materials); flat pattern unfolded from the 3D model with the k-factor of the Design rules. Not a finite-element forming simulation: thinning, wrinkling and exact springback need a CAE run for critical parts.

Notes: 1. Remove all sharp edges 0.2 x 45°. Do not scale the drawing. The 3D model is the master.

Design to Cost
Drawing no.
EX26-0001-SIM
Rev **A**

Part	EX26-0001	**Material**	Galvanised steel GI
Revision	A	**Thickness**	2 mm

General tolerance	ISO 2768-mK
Projection	
Scale	-
Sheet	1 / 1
Drawn	Amit Shukla · 2026-10-26
Checked	-
Approved	-